Data Quality Report

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1. Introduction

This dataset is sourced from cal-wildfires-23224143.csv and contains 10,000 records. The analysis examines the dataset's distribution, incorporating definitions from the Natural Resources Website and the Damage Inspection (DINS) Database Dictionary to ensure accurate interpretation of each field. Before further analysis, Preliminary Data Processing is conducted to refine the dataset. Once cleaned, an Analytical Base Table (ABT) and visualizations illustrating the distributions of both continuous and categorical features are provided in the appendix. This report primarily focuses on identifying data quality issues and proposing strategies to address them, ensuring a high-quality, reliable dataset for further analysis.

2. Preliminary Data Processing

2-1 Duplicate Removal

While no fully duplicate records exist due to the uniqueness of the "OBJECTID" field, 217 records were removed based on subset definitions established below. These definitions were derived from field distributions and dataset characteristics to ensure a meaningful approach to handling duplicates.

Two subset definitions were used:

- **Subset 1:** Records with identical values in key structural and damage-related fields (e.g., Damage, Street Name, County, Site Address (parcel), Structure Type, Structure Category, Roof Construction, etc.) were considered duplicates, leading to the removal of 196 records.
- **Subset 2:** Similar to Subset 1, but with "County" and "Site Address (parcel)" replaced by "Street Type (e.g., road, drive, lane, etc.)". This subset primarily contained missing values, leading to the removal of an additional 21 records.

2-2 Constant Feature Removal

The "State" and "Hazard Type" features were removed as they contained only a single value ('CA' and 'Fire', respectively), providing no additional information.

2-3 Feature Type Conversion

To ensure consistent data representation, the following transformations are applied:

1. **# Units in Structure (if multi unit)**, **Assessed Improved Value (parcel)**, **Year Built (parcel)**, **Latitude**, and **Longitude** are converted to integer or float, as they represent numerical values.
2. **OBJECTID** is treated as an object type since it serves as a unique identifier for each record.
3. **Incident Start Date** is converted to datetime format to retain more information and facilitate time-based analysis.
4. All remaining features are converted to categories for better data representation and efficiency. Although **ZIP CODE** and *** Street Number** contain only numbers, they do not represent meaningful numerical values for calculations like mean or median. Therefore, they are also converted to categorical data.

2-4 Final Dataset Summary

After processing, the refined dataset consists of 9,783 records, with 6 continuous features and 25 categorical features. The ABT and visualizations in this report are based on this refined dataset.

3. Issues Identified and Possible Solutions

3-1 Continuous Features

Issue 1 - Missing Values

Problematic Fields

- **# Units in Structure (if multi unit)** (69.63%)
- **Assessed Improved Value (parcel)** (4.83%)
- **Year Built (parcel)** (24.26%)

Possible solutions

For features with a missing rate below 30%, impute missing values with reasonable estimates such as median, mode, or average. For features with a missing rate higher than 60%, consider removing them as they may provide limited analytical value.

Issue 2 - Invalid Data

Problematic Fields

- **Year Built (parcel):** Contains 758 records with a value of 0, which is incorrect.

Possible solutions

Treat invalid data as missing and apply imputation techniques used for handling missing values.

Issue 3 - Skewed Distributions & Outliers

Problematic Fields

- **# Units in Structure (if multi unit):** Highly right-skewed, with over 90% of non-missing values being 0, resulting in all non-zero values outliers.
- **Assessed Improved Value (parcel):** Highly right-skewed with a significant number of extreme values, as observed in the box plot and high standard deviation.

Possible solutions

Investigate whether extreme values are due to data entry errors. When imputing missing values, avoid using values that may be affected by outliers. For outliers, assess whether an appropriate threshold can be applied for clamping to mitigate their impact.

Further Insights

Latitude and **Longitude** have no missing values and contain reasonable data points. Both exhibit multimodal distributions (**Latitude** concentrated around 39-40 and 34 while **Longitude** concentrated around -122 and -128) Since these features are complete and valid, they could be leveraged to infer missing address-related information (e.g., ZIP code, region) or serve as a reference for geolocation-based analyses.

3-2 Categorical Features

Issue 1 - Missing Values

Problematic Fields

- Low to moderate Missing rates (<30%)
    - *** Street Number** (3.50%)**, * Street Name** (4.34%)**, * Street Type (e.g. road, drive, lane, etc.)** (10.85%)**, * City** (23.90%)**, County** (0.03%)**, * Roof Construction** (0.49%)**, * Eaves** (0.96%)**, * Vent Screen** (0.96%)**, * Exterior Siding** (0.75%)**, * Window Pane** (0.75%)**, * Fence Attached to Structure** (16.61%), **Site Address (parcel)** (3.55%)
- Moderate Missing rates (30 - 70%)
    - **Zip Code** (38.92%)**, Community** (58.15%)
- Extreme missing rates (>70%)
    - **Structure Defense Actions Taken** (72.87%)**, Distance - Propane Tank to Structure** (81.86%)**, Distance - Residence to Utility/Misc Structure > 120 SQFT** (85.41%)

Possible Solutions

Since **Site Address (parcel)** comprises **Street Number**, **Street Name**, **Street Type**, **City, State**, and **Zip Code**, it can be used to impute missing values for these location-based fields. However, once the necessary data is extracted, this feature should be removed to reduce redundancy. For other features, missing values should be imputed using the mode or labeled as “Unknown” if the category exists to enhance consistency, while features with excessive missing data should be dropped.

Issue 2 - Invalid Data

Problematic Fields

- Contain 0: *** Street Number, Zip Code**
- Contain blank value: *** Roof Construction, * Eaves, * Vent Screen, * Exterior Siding, * Window Pane, * Deck/Porch On Grade, * Deck/Porch Elevated, * Patio Cover/Carport Attached to Structure, Distance - Residence to Utility/Misc Structure > 120 SQFT**
- Others: *** Street Name** (Includes only numbers)

Possible Solutions

Treat invalid data as missing and apply imputation techniques used for handling missing values.

Issue 3 - Inconsistent Data Formatting

Problematic Fields

- *** Street Type (e.g. road, drive, lane, etc.):** Variations such as "Highway" vs. "Hwy" create inconsistencies.
- **Community:** "Paradise" appears as two separate categories due to trailing spaces.
- **Incident Start Date:** Contains timestamps despite being labeled as a date field, with most showing "12:00:00 AM."
- *** CAL FIRE Unit:** “MVU” is the formal name of “SDU.”
- *** City**: Some cities appear as separate categories due to inconsistent cases and trailing spaces, like “Colfax” and “Colfax “.
- *** Exterior Siding:** Variations in formatting cause similar values to be treated as distinct categories, such as "Stucco Brick Cement" and "Stucco/Brick/Cement".
- *** Structure Type**: A typographical error results in "Single Famliy Residence Single Story" being incorrectly treated as a separate category from "Single Family Residence Single Story".
- **Structure Category**: 8 records of “Multi Family Residence Multi Story” structure types are incorrectly categorized under “Single Residence” structure category.

Possible Solutions

Remove trailing spaces across all features and standardize naming conventions while accounting for differences in uppercase and lowercase variations. For **Street Type**, consider merging rare categories into "Other" to reduce cardinality. For **Incident Start Date**, timestamps should either be removed or stored in a separate column.

Issue 4 - Mixed Granularity

Problematic Fields

- *** Eaves:** can be structured into “Applicable” vs. “Not Applicable”, with “Applicable” further divided into “Has Eaves” and “No Eaves.” If classified as “Has Eaves,” it can be further specified as “Unenclosed” or “Enclosed.”
- *** Vent Screen:** can be categorized into "Has Vent" vs. "No Vent", with “Has Vent” further divided into “Screened” and “Unscreened.” If classified as “Screened,” it can be further specified as “Mesh Screen > 1/8"""” or “Mesh Screen <= 1/8"""”
- *** Patio Cover/Carport Attached to Structure:** can be restructured into “Has Patio Cover/Carport” vs. “No Patio Cover/Carport”, with “Has Patio Cover/Carport” further divided into “Combustible” and “Non-Combustible.”
- *** Fence Attached to Structure:** can be restructured into “Has Fence” vs. “No Fence”, with “Has Fence” further divided into “Combustible” and “Non-Combustible.”
- *** Deck/Porch On Grade:** can be restructured into “Has Deck/Porch” vs. “No Deck/Porch”, with “Has Deck/Porch” further divided into different materials.
- *** Deck/Porch Elevated:** can be restructured into “Has Deck/Porch” vs. “No Deck/Porch”, with “Has Deck/Porch” further divided into different materials.

Possible Solutions

Consider splitting these features into multiple fields to maintain proper granularity.

Issue 5 - High cardinality

Problematic Fields

- *** Street Number** (5578)**, * Street Name** (3956)**, Site Address (parcel)** (8508): These fields contain high cardinality with over 3,000 unique values.

Possible Solutions

Consider dropping these fields, as they provide little predictive power in modeling.

Issue 6 - Long-Tail Categories

Problematic Fields

- *** Street Type (e.g. road, drive, lane, etc.):** There are 19 categories, but the top two account for more than 50% of the data, while over half of the categories individually contribute no more than 3%.
- *** Structure Type**: With a cardinality of 19, the top three categories make up over 85% of the data, while each of the remaining categories contributes no more than 5% individually.
- **Community :** There are 317 categories, but almost 300 of them have fewer than 1% of the total records.
- *** CAL FIRE Unit:** There are 27 categories, but half have fewer than 1% of the total records.
- *** City:** The dataset contains 252 unique city names, yet around 200 of them have fewer than 1% of the total records.
- **Structure Category, * Deck/Porch On Grade, * Deck/Porch Elevated:** Some categories contribute fewer than 1% of the total records.

Possible Solutions

Merge categories under a more general name or consolidate those below a defined threshold (e.g., <1%) into an 'Other' category to reduce feature dimensionality where applicable.

Further Insights

In **Incident Start Date**, 24.39% of the records share the timestamp **1/7/2025 8:02:01 AM**, which requires further investigation to determine whether it corresponds to a major wildfire event.

4. Conclusion

The main challenges in this dataset include **Missing Values**, **Skewed Distributions & Outliers**, **Invalid Data**, **Inconsistent Data Formatting**, **Mixed Granularity**, **High cardinality**, and **Long-Tail Categories**. When handling missing values, the appropriate approach may vary depending on the characteristics of each field. Notably, treating invalid data and entries labeled as "Unknown" as uncertain values can significantly increase the level of uncertainty in many fields. If the proportion of uncertain data becomes excessively high (e.g., exceeding 80%), removing the feature is often the preferable choice. Additionally, trailing space issues were observed in several fields, it is recommended to apply trailing space removal across all categorical fields to minimize potential inconsistencies during data processing.

In conclusion, all the features require further investigation to see the best startegy to address all the issues. The actual action and the reason why that action is considered better than other options will be delivered in the Data Quality Plan.

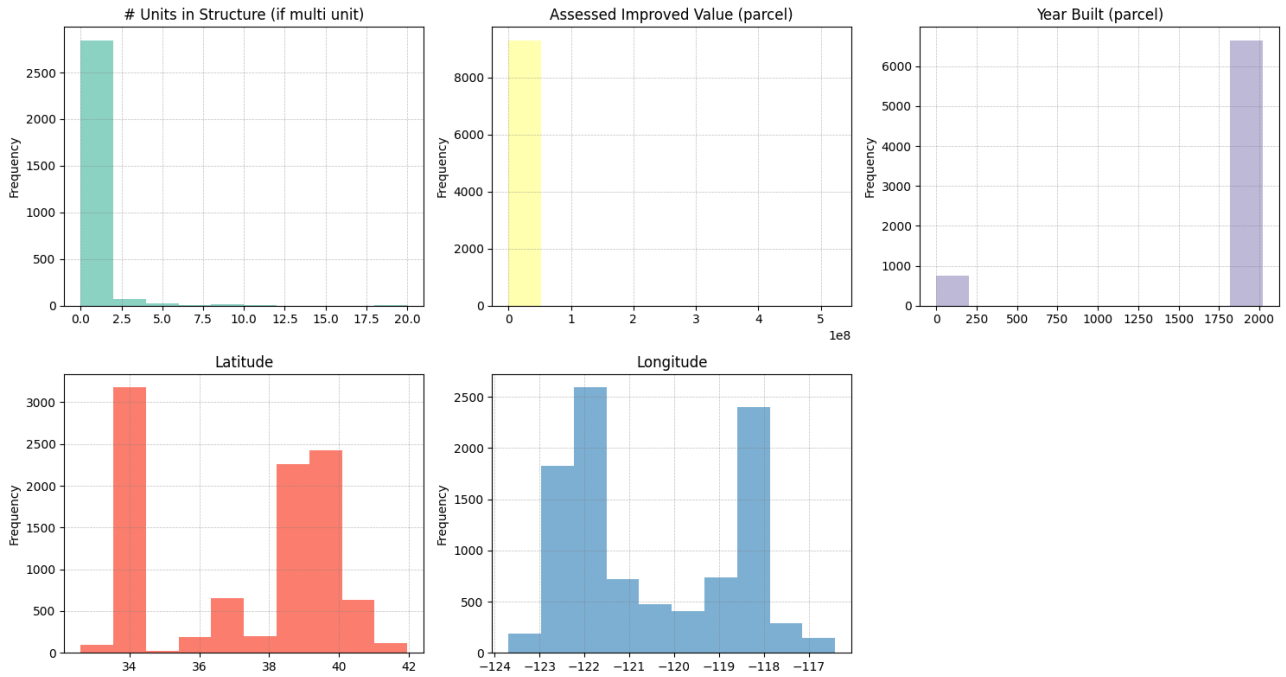
5. Appendix

5-1 Continuous Features

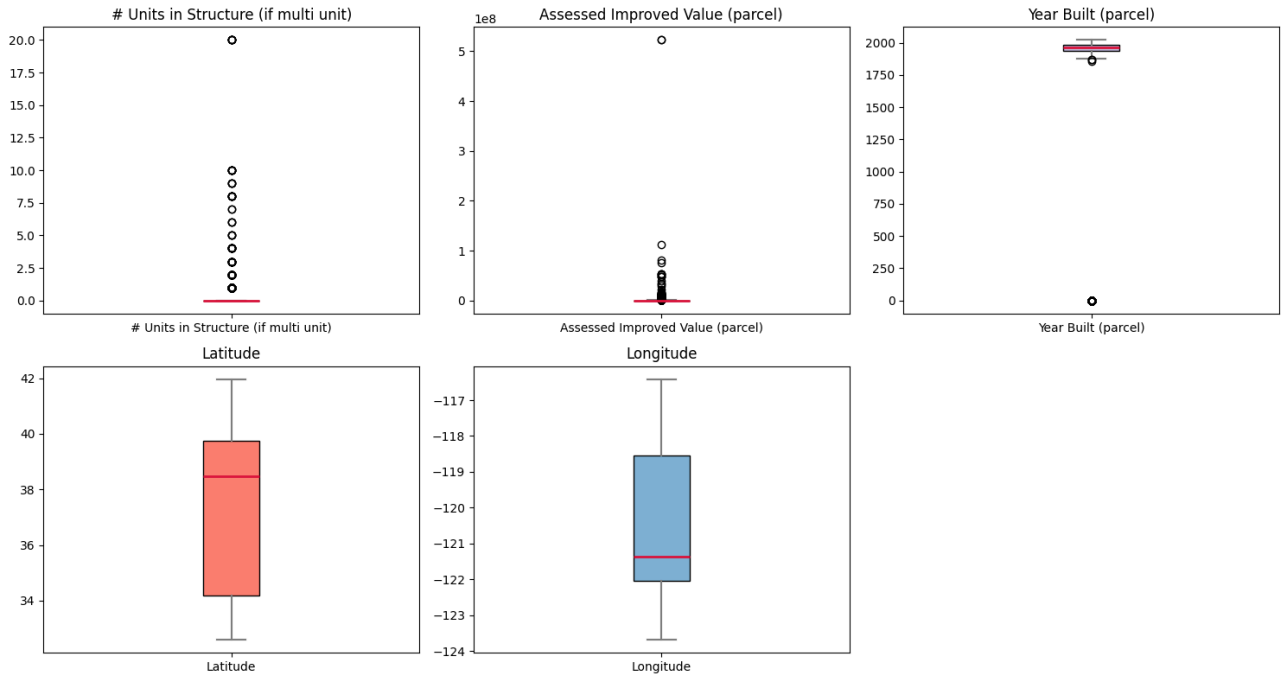
Dataset ABT

	count	missing%	card	min	25%	mean	50%	75%	max	std
# Units in Structure (if multi unit)	2971.0	69.63	12	0.0	0.0	0.25	0.0	0.0	20.0	1.35
Assessed Improved Value (parcel)	9310.0	4.83	7636	0.0	69501.0	551674.88	159135.0	334685.25	522652568.0	8027409.24
Year Built (parcel)	7410.0	24.26	135	0.0	1940.0	1763.77	1961.0	1982.0	2022.0	595.96
Latitude	9783.0	0.00	9779	32.59	34.19	37.33	38.47	39.74	41.96	2.5
Longitude	9783.0	0.00	9782	-123.69	-122.05	-120.48	-121.36	-118.54	-116.42	1.81

Histogram



Boxplot



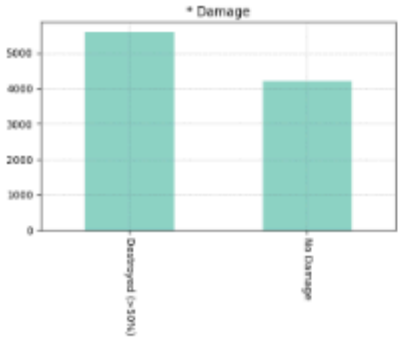
5-2 Categorical Features

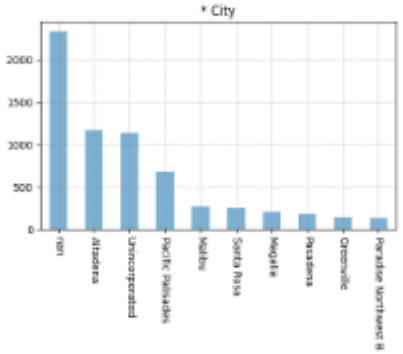
Dataset ABT

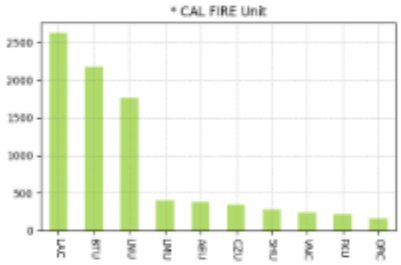
Feature	count	missing%	card	mode	mode_freq	mode%	2nd_mode	2nd_mode_freq	2nd_mode%
* Damage	9783	0.00	2	Destroyed (>50%)	5587	57.11	No Damage	4196	42.89
* Street Number	9441.0	3.50	5578.0	0	503	5.14	NaN	342	3.5
* Street Name	9358	4.34	3956	NaN	425	4.34	Pacific Coast	84	0.86
* Street Type (e.g. road, drive, lane, etc.)	8722	10.85	19	Road	3390	34.65	Drive	1638	16.74
* City	7445	23.90	252	NaN	2338	23.9	Altadena	1171	11.97
Zip Code	5975.0	38.92	155.0	NaN	3808	38.92	0	1866	19.07
* CAL FIRE Unit	9783	0.00	27	LAC	2629	26.87	BTU	2176	22.24
County	9780	0.03	47	Los Angeles	2629	26.87	Butte	2176	22.24
Community	4094	58.15	416	NaN	5689	58.15	Paradise	582	5.95
Incident Start Date	9783	0.00	NaN	2025-01-07 08:02:01	2386	24.39	2018-11-08 00:00:00	1922	19.65
Structure Defense Actions Taken	2654	72.87	10	NaN	7129	72.87	Unknown	2079	21.25
* Structure Type	9783	0.00	19	Single Family Residence Single Story	3534	36.12	Utility Misc Structure	2639	26.98
Structure Category	9783	0.00	7	Single Residence	6567	67.13	Other Minor Structure	2641	27.0
* Roof Construction	9735	0.49	10	Asphalt	4449	45.48	Metal	1502	15.35
* Eaves	9689	0.96	6	Unknown	3951	40.39	Unenclosed	3216	32.87
* Vent Screen	9689	0.96	7	Unknown	2982	30.48	Mesh Screen > 1/8""	2565	26.22
* Exterior Siding	9710	0.75	11	Stucco Brick Cement	2629	26.87	Wood	2183	22.31
* Window Pane	9710	0.75	5	Multi Pane	3163	32.33	Single Pane	2882	29.46
* Deck/Porch On Grade	9783	0.00	6	Masonry/Concrete	2998	30.64	No Deck/Porch	2928	29.93
* Deck/Porch Elevated	9783	0.00	6	No Deck/Porch	4675	47.79		1626	16.62
* Patio Cover/Carport Attached to Structure	9783	0.00	5	No Patio Cover/Carport	4247	43.41	Combustible	2012	20.57
* Fence Attached to Structure	8158	16.61	4	No Fence	4142	42.34	Non Combustible	1736	17.75
Distance - Propane Tank to Structure	1775	81.86	6	NaN	8008	81.86	Unknown	492	5.03
Distance - Residence to Utility/Misc Structure > 120 SQFT	1427	85.41	6	NaN	8356	85.41	<30'	770	7.87
Site Address (parcel)	9436	3.55	8508	NaN	347	3.55		143	1.46

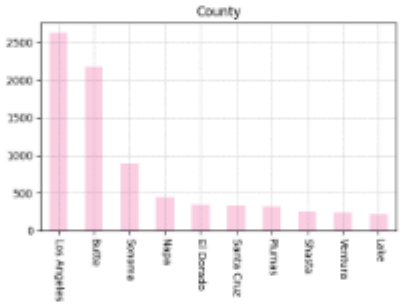
Barplots

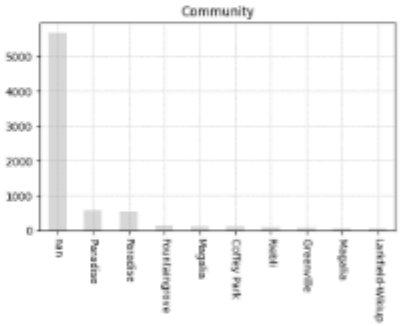
Since some categories have excessively high cardinality, for better readability, the following bar plots display only the top 10 values for each feature.

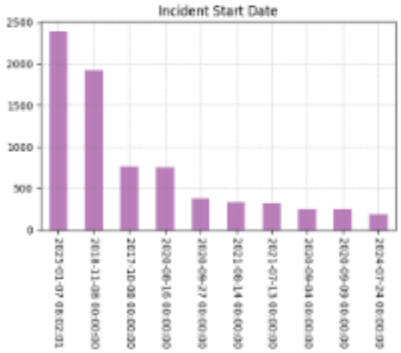


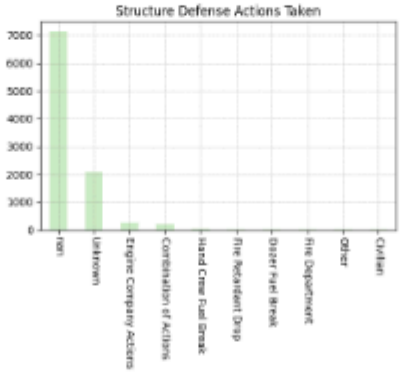


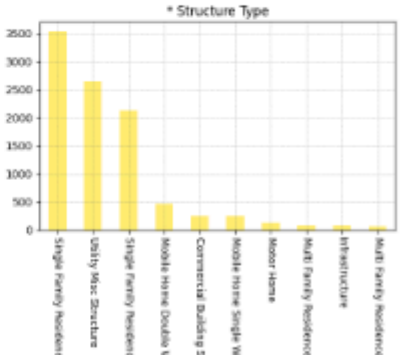








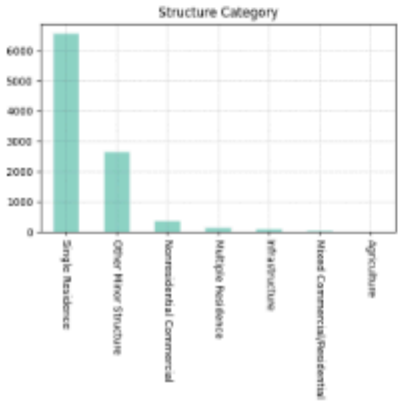


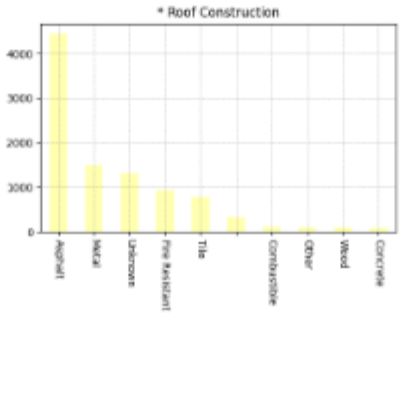


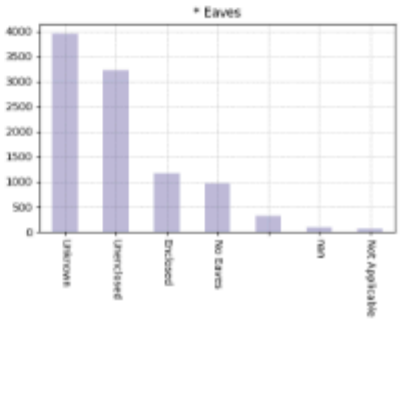
Structure Category

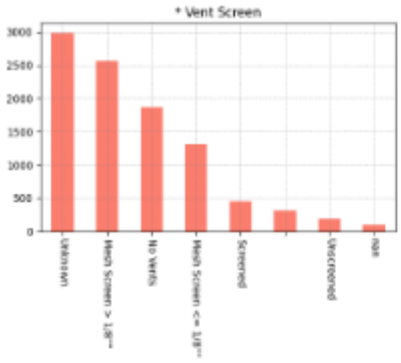
* Roof Construction

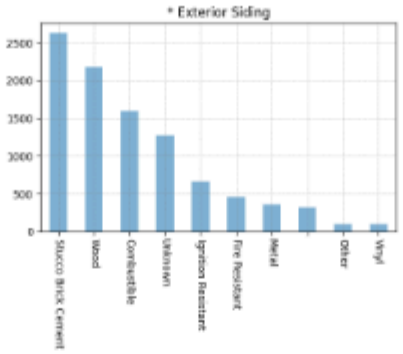
* Stories

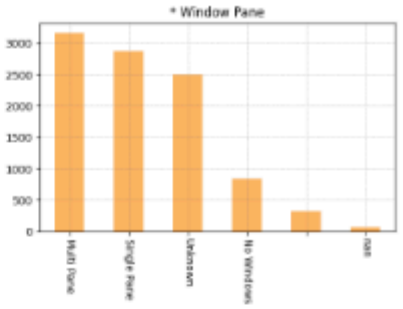

Structure Category
Single Residence
Other Minor Structure
Nonresidential Commercial
Multiple Residence
Infrastructure
Mixed Commercial/Residential
Agriculture

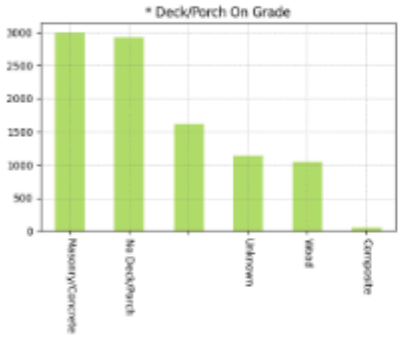

* Roof Construction
Asphalt
Metal
Unknown
Fire Resistant
Tile
Combustible
Other
Wood
Concrete

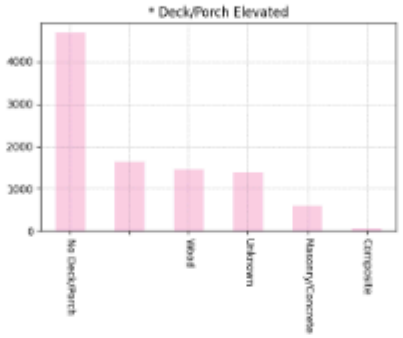

* Eaves
Unknown
Unenclosed
Enclosed
No Eaves
nan
Not Applicable

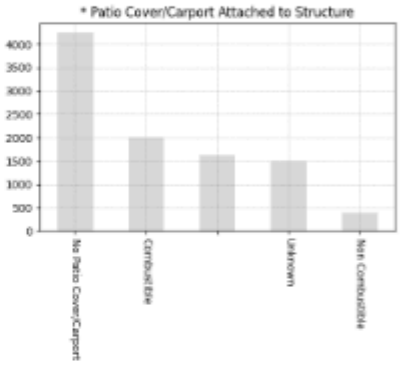

* Vent Screen
Unknown
Mesh Screen > 1/8"
No Vents
Mesh Screen <= 1/8"
Screened
Unscreened
nan

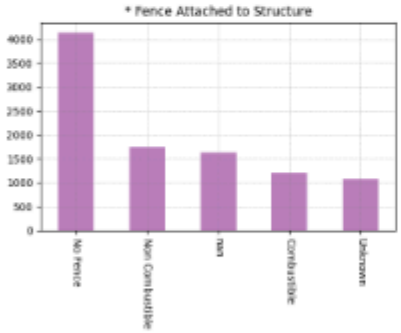

* Exterior Siding
Stucco Brick Cement
Wood
Combustible
Unknown
Ignition Resistant
Fire Resistant
Metal
Other
Vinyl

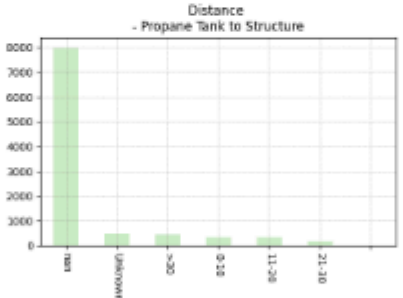

* Window Pane
Multi Pane
Single Pane
Unknown
No Windows
nan

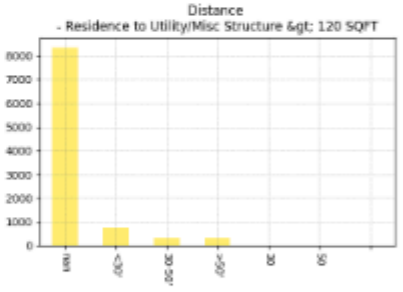

* Deck/Porch On Grade
Masonry/Concrete
No Deck/Porch
Unknown
Wood
Composite

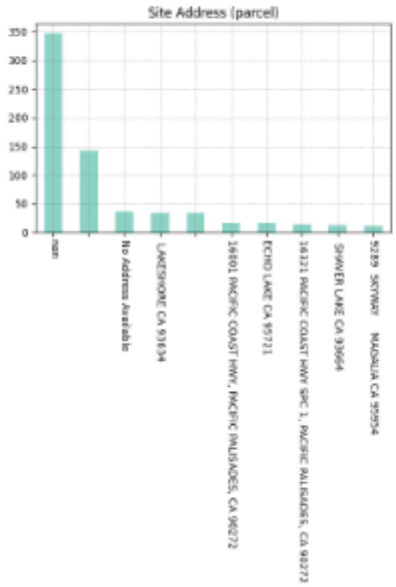

* Deck/Porch Elevated
No Deck/Porch
Wood
Unknown
Masonry/Concrete
Composite


* Patio Cover/Carport Attached to Structure
No Patio Cover/Carport
Combustible
Unknown
Non Combustible


* Fence Attached to Structure
No Fence
Non Combustible
nan
Combustible
Unknown


Distance
- Propane Tank to Structure
nan
Unknown
>30
0-10
11-20
21-30


Distance
- Residence to Utility/Misc Structure > 120 SQFT
nan
<30'
30-50'
>50'
30
50


Site Address (parcel)
350
300
250
200
150
100
50
0
nan
No Address Available
LAKESHORE CA 93634
14601 PACIFIC COAST HWY, PACIFIC PALISADES, CA 90272
ECHO LAKE CA 95721
14321 PACIFIC COAST HWY SPC 1, PACIFIC PALISADES, CA 90272
SHAVER LAKE CA 93664
5289 SKYWAY MAGALIA CA 95954